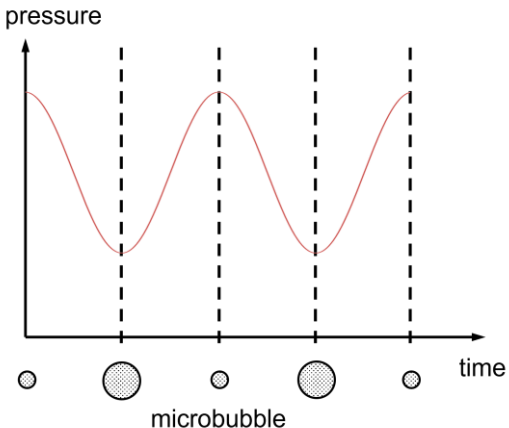


Qn	Suggested Solutions
8(a)	$\text{unit of } Z = (\text{unit of } \rho)^a (\text{unit of } \kappa)^b$ $\text{kg m}^{-2} \text{ s}^{-1} = (\text{kg m}^{-3})^a (\text{m kg}^{-1} \text{ s}^2)^b$ $= \text{kg}^{a-b} \text{ m}^{-3a+b} \text{ s}^{2b}$
	Comparing the powers, $2b = -1$ $b = -\frac{1}{2}$ $a - b = 1$ $a = 1 - \frac{1}{2}$ $= \frac{1}{2}$
(b)(i)	$\frac{I_r}{I_i} = \frac{(Z_{\text{mus}} - Z_{\text{bone}})^2}{(Z_{\text{mus}} + Z_{\text{bone}})^2}$ $= \frac{(1.71 - 7.63)^2}{(1.71 + 7.63)^2}$
	$\frac{I_r}{I_i} = 0.402$
(ii)	A significant fraction of the ultrasound beam will not be able to reach the heart as 40% of it gets reflected by the ribs.
(iii)	Liver and kidney
	(It is difficult to detect a boundary if the ultrasound is not reflected from the boundary. $\frac{I_r}{I_i}$ is a minimum when the impedance of the two tissues is as small as possible. i.e. when $Z_1 - Z_2$ is smallest since denominator are similar order of magnitude.)
(c)(i)	$\frac{I}{I_0} = e^{-\mu f x}$ $\frac{1}{2} = e^{-0.23(5.0)x}$
	$x = 0.60 \text{ cm}$
(ii)	Based on the formula, $I = I_0 e^{-\mu f x}$ For I to be constant, f must decrease for x to increase OR To ensure that the ultrasound beam has a minimum intensity for detection when it enters deeper into the body, it must be attenuated less / its intensity should decrease at a slower rate.
	Low frequency ultrasound waves should be used.

(d)(i)	$\frac{I_t}{I_i} = \frac{4Z_{\text{PZT}}Z_{\text{skin}}}{(Z_{\text{PZT}} + Z_{\text{skin}})^2}$ $= \frac{4(30 \times 10^5)(1.7 \times 10^5)}{(30 \times 10^5 + 1.7 \times 10^5)^2}$
	$\frac{I_t}{I_i} \times 100\% = 0.203 \times 100\%$ $= 20.3\%$ $\approx 20\% \text{ (Shown)}$
(ii)	$Z_{\text{matching layer}} = \sqrt{Z_{\text{PZT}}Z_{\text{skin}}}$ $= \sqrt{(30 \times 10^5)(1.7 \times 10^5)}$ $= 7.14 \times 10^5 \text{ g cm}^{-2} \text{ s}^{-1}$
	$\frac{I_t}{I_i} = \frac{4Z_{\text{PZT}}Z_{\text{matching layer}}}{(Z_{\text{PZT}} + Z_{\text{matching layer}})^2}$ $= \frac{4(30 \times 10^5)(7.14 \times 10^5)}{(30 \times 10^5 + 7.14 \times 10^5)^2}$ $= 0.621$
	$\frac{I_t}{I_i} = \frac{4Z_{\text{matching layer}}Z_{\text{skin}}}{(Z_{\text{matching layer}} + Z_{\text{skin}})^2}$ $= \frac{4(7.14 \times 10^5)(1.7 \times 10^5)}{(7.14 \times 10^5 + 1.7 \times 10^5)^2}$ $= 0.621$
	(Since $P = IA$, efficiency is proportional to the ratio of intensities.) $\text{efficiency} = 0.621^2 \times 100\%$ $= 38.6\%$ $\approx 39\%$
(iii)	$\text{thickness} = \frac{1}{4} \lambda$ $= \frac{1}{4} \left(\frac{v}{f} \right)$ $= \frac{1}{4} \left(\frac{2500}{5.0 \times 10^6} \right)$ $= 1.25 \times 10^{-4} \text{ m}$

(e)	 pressure time microbubble
(f)	Ionising radiation interacts with water in the cells and break the bonds that hold the water molecule together, producing H and OH free radicals.
	Free radicals combine to form toxic substances like H_2O_2 which can destroy the cell.